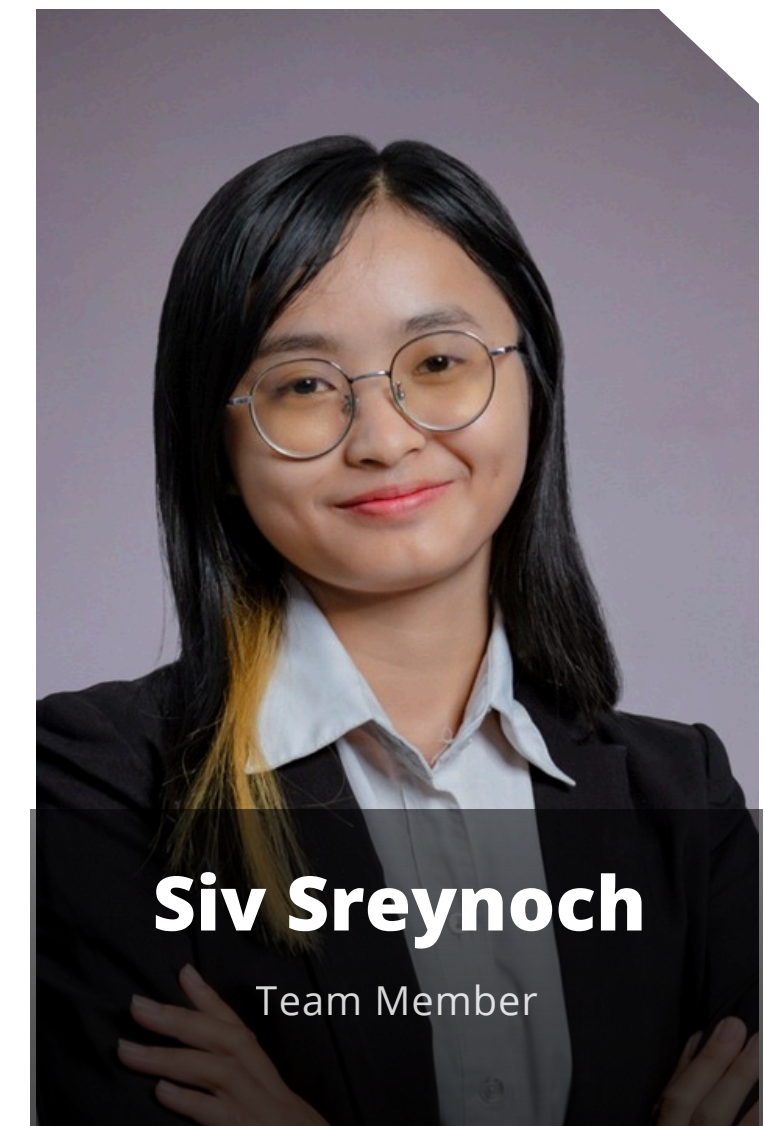
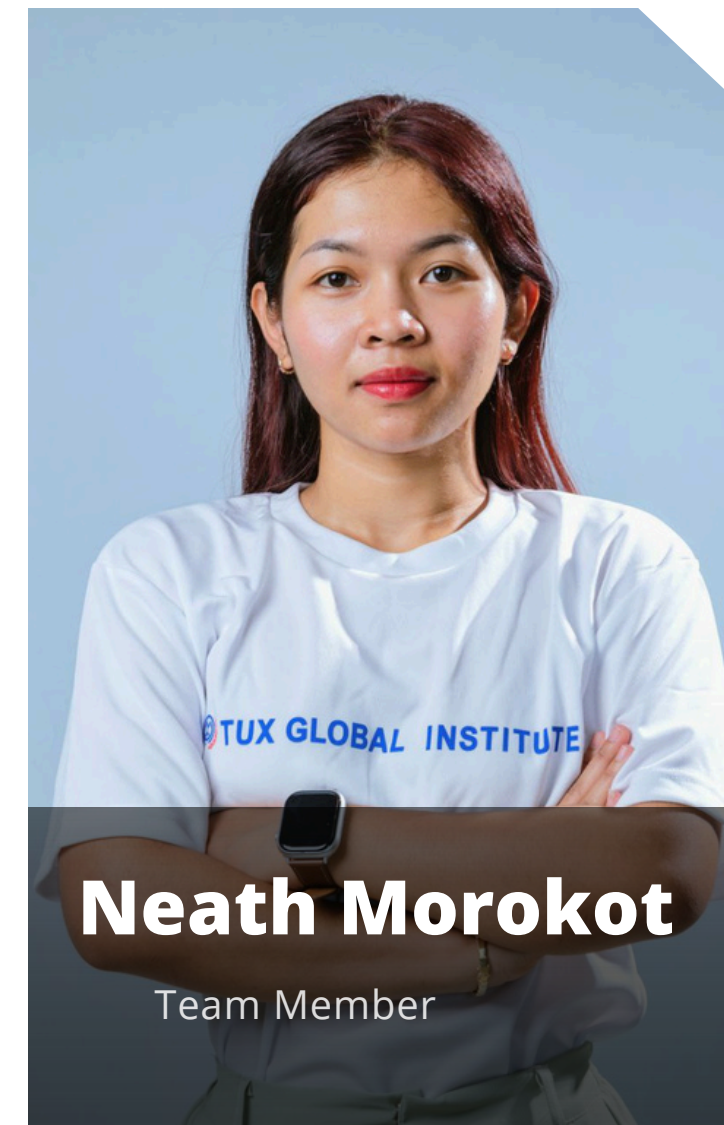
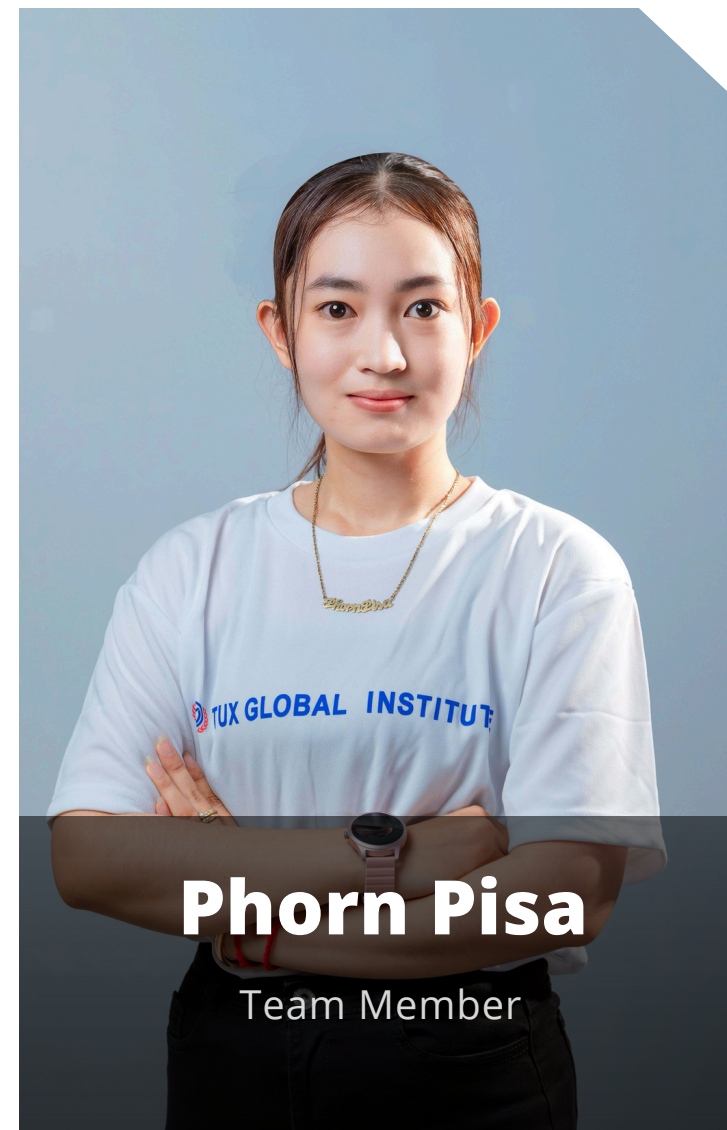


Implementation of Secure Access Controls **on the File Server**

Vanna Sarem

Windows Security for Enterprise Organizations





Introduction

- This project focus on implementing the Folder name HQ on the file server name “**SRVF.testnet**” for everyone access to get read the announcement form HR department.
- The goal is to **make all staffs can read all files** that HR department put in this folder and make sure the **Internship staff cannot read**. Moreover, HQ store **Raid5**, set **10GB Disk Quota** for everyone and **setup backup** schedule for everyday at the mid night

Requirements

Servers

- Domain Controller (DC): Manages domain test.net, authentication, and group policies.
- File Server (FS): Named srvf, stores department folders and enforces access control.

Operating System

- Windows Server 2025
- Active Directory (AD)
- Create OUs: HQ, HR, IT, Internship with sub-OUs Users and Computers.
- Create Security Groups: Staff, Intern.

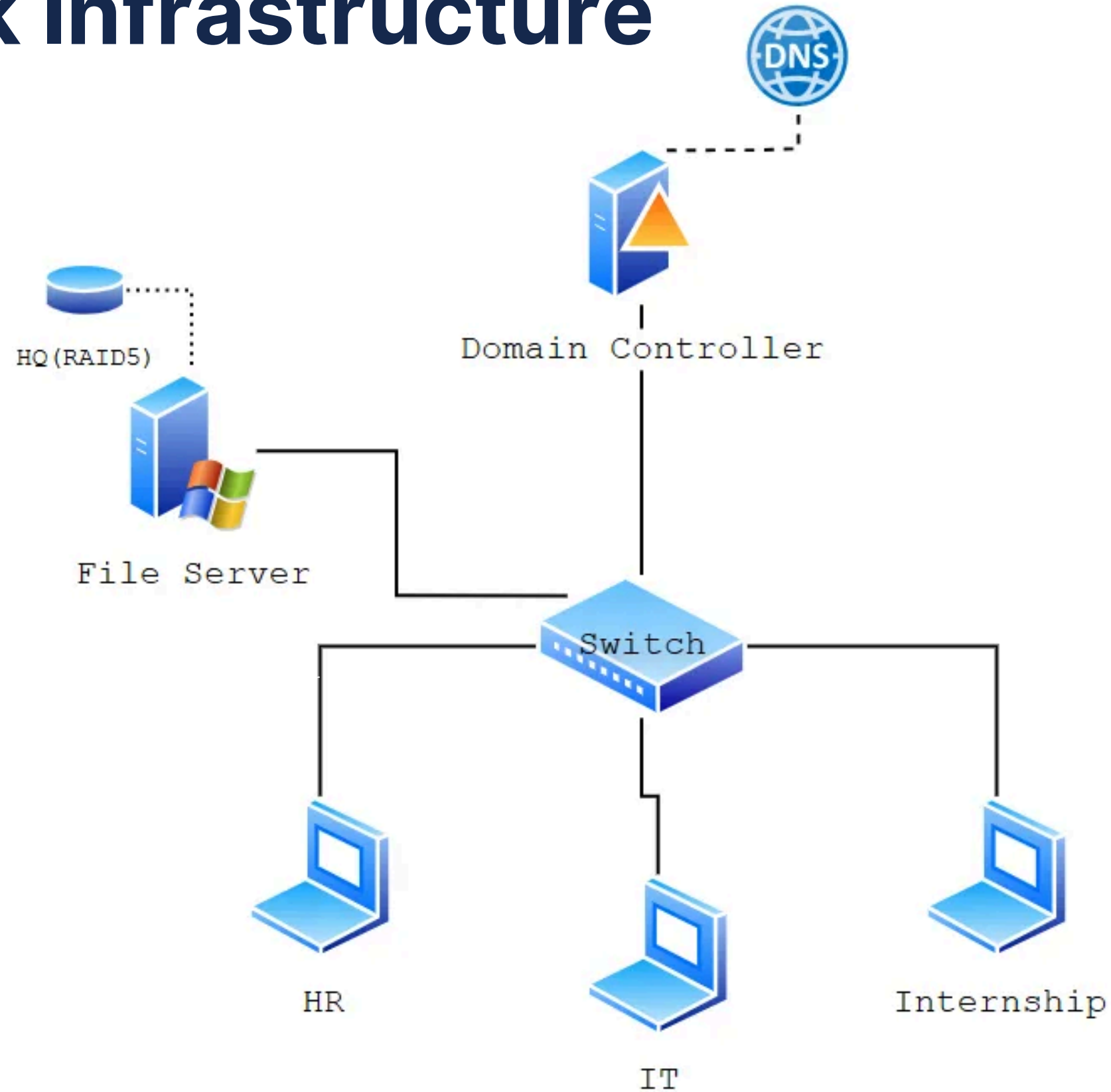
• User & Group Policies

- Staff group → Full read access to shared folders.
- Intern group → Restricted access, no access to certain folders.

• File Server Configuration

- Configure RAID 5 for HQ storage.
- Set 10 GB disk quota per user.
- Schedule daily midnight backups.
- Shared Folder: Announcement
 - HR: Full Control
 - Staff: Read-Only
 - Interns: No Access
 - Admins: Full Control

Network Infrastructure



Domain Controller (test.net)	192.168.1.2
DNS	192.168.1.2
File Server (srvf.test.net)	192.168.1.3
HR	192.168.1.100
IT	192.168.1.110
Internship	192.168.1.120

Task 1: Enviroment Set Up the Domain Controller (DC)

1.1 Install Domain Controller (test.net)

- Install Windows Server.
- Open Server Manager → click Add roles and features.
- Select Active Directory Domain Services (AD DS) → Next → install.
- After installation, click the notification flag → Promote this server to a domain controller.
- Choose Add a new forest → enter test.net as root domain name.
- Set Directory Services Restore Mode (DSRM) password → complete the wizard and reboot.

Active Directory Domain Services Configuration Wizard

Deployment Configuration

TARGET SERVER
DC

Deployment Configuration

Select the deployment operation

- ☐ Add a domain controller to an existing domain
- ☐ Add a new domain to an existing forest
- ☒ Add a new forest

Specify the domain information for this operation

Root domain name:

More about deployment configurations

< Previous Next > Install Cancel

Active Directory Domain Services Configuration Wizard

Domain Controller Options

TARGET SERVER
DC

Deployment Configuration

Domain Controller Options

Select functional level of the new forest and root domain

Forest functional level:

Domain functional level:

Specify domain controller capabilities

- ☒ Domain Name System (DNS) server
- ☒ Global Catalog (GC)
- ☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password:

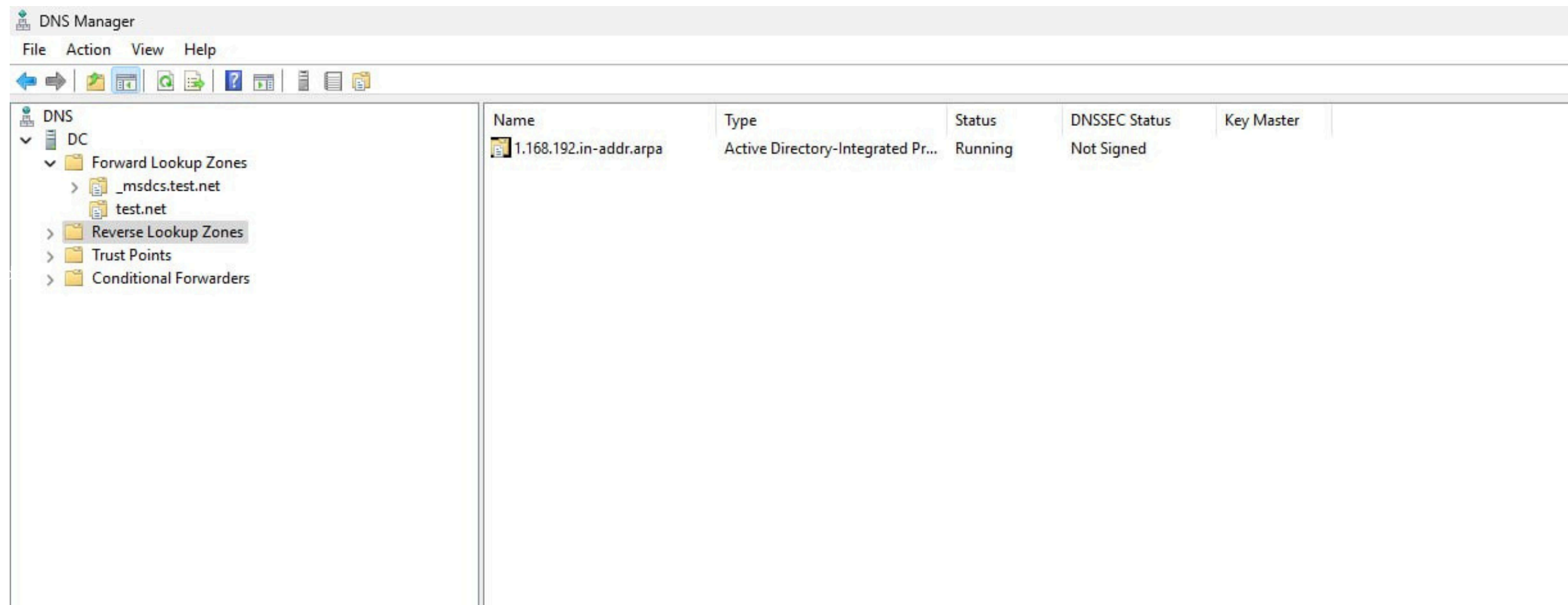
Confirm password:

More about domain controller options

Task 1: Enviroment Set Up the Domain Controller (DC)

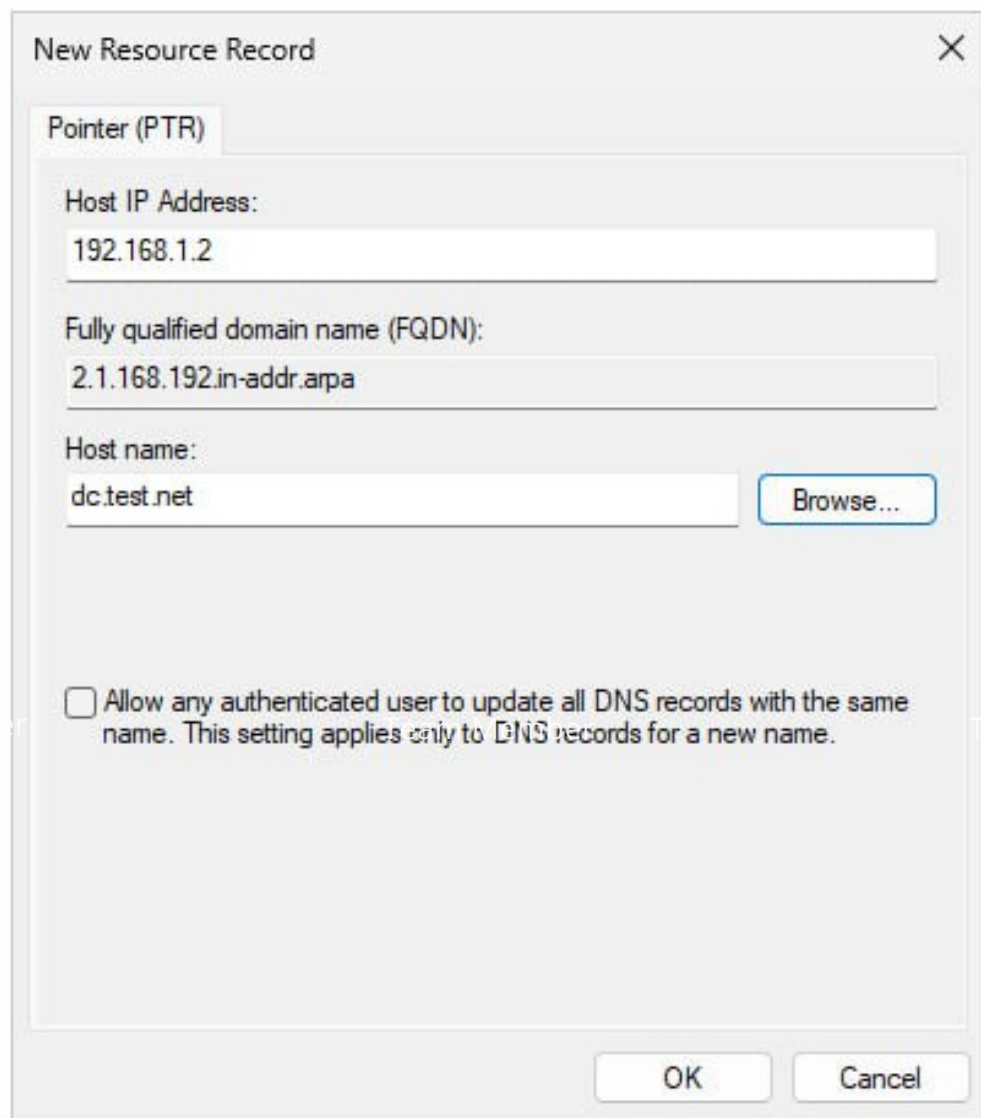
1.2 Configure DNS for AD

- During AD installation, DNS will install automatically.
- Confirm DNS setup:
- Open DNS Manager → check forward lookup zone → ensure test.net zone exists.
- Verify the domain controller's IP is listed as DNS server.



Task 1: Enviroment Set Up the Domain Controller (DC)

- Add a pointer
- In cmd type **nslookup** to query the Domain Name System



New Resource Record

Pointer (PTR)

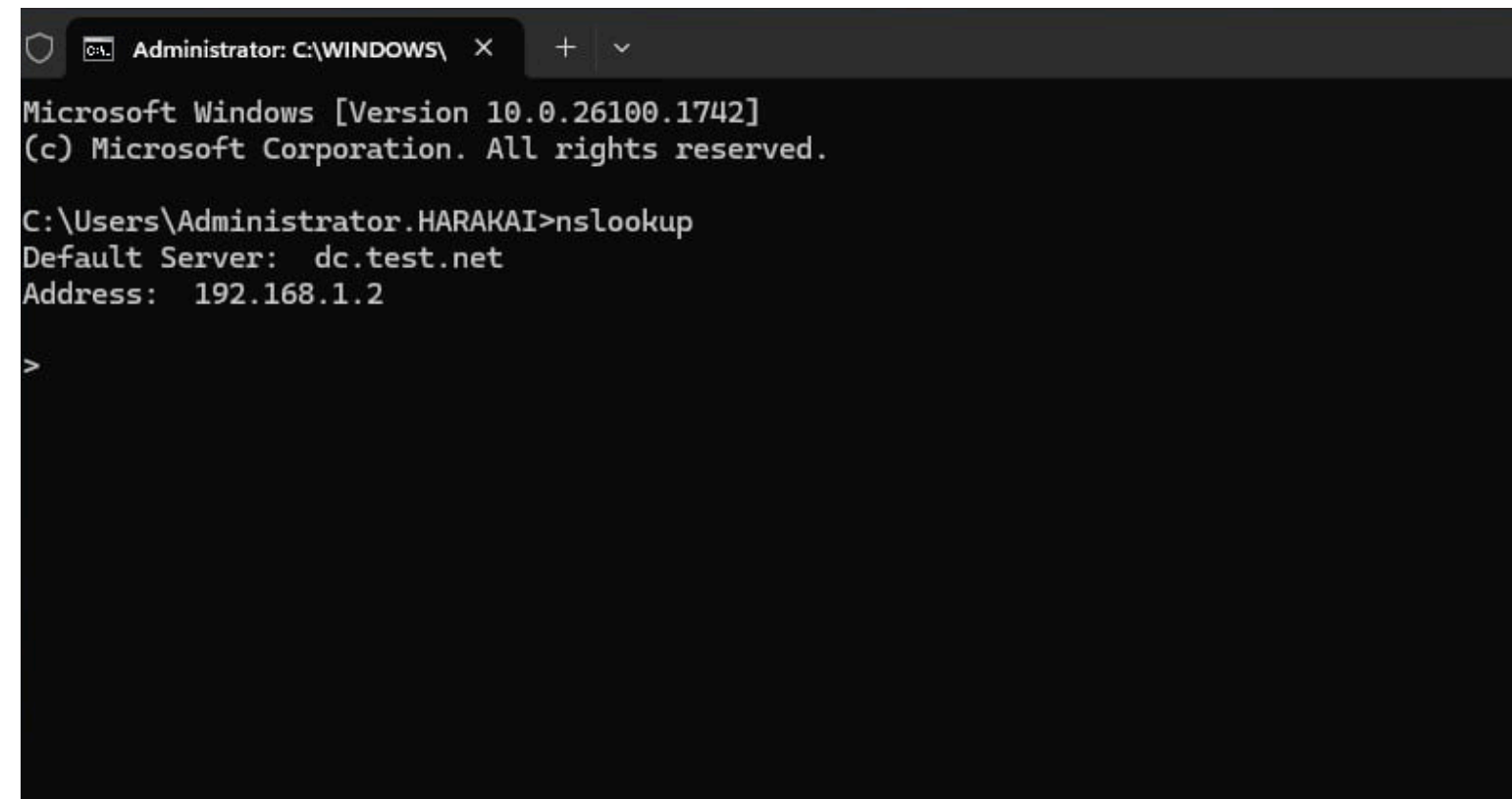
Host IP Address:
192.168.1.2

Fully qualified domain name (FQDN):
2.1.168.192.in-addr.arpa

Host name:
dc.test.net Browse...

☐ Allow any authenticated user to update all DNS records with the same name. This setting applies only to DNS records for a new name.

OK Cancel



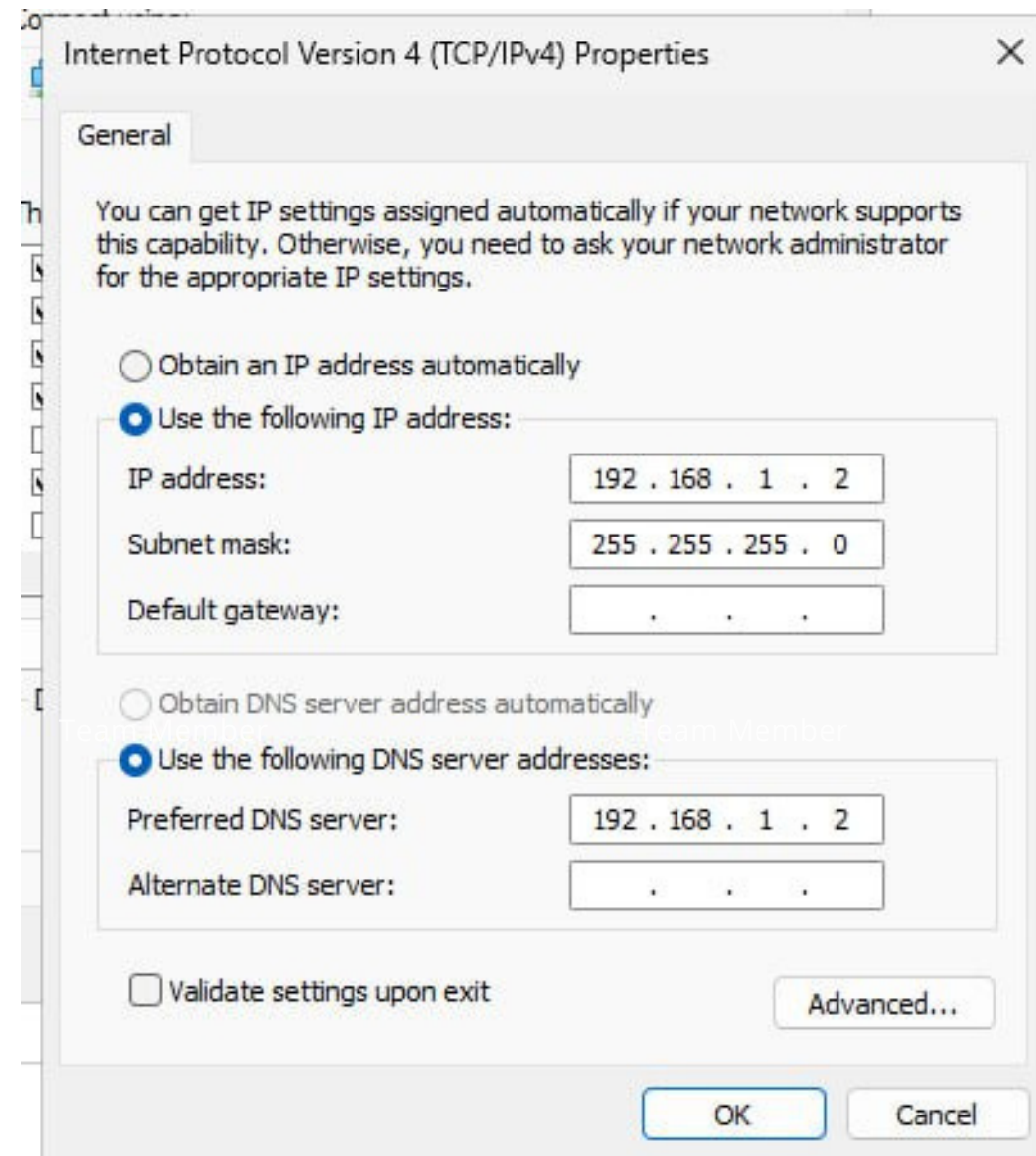
```
Administrator: C:\WINDOWS\
Microsoft Windows [Version 10.0.26100.1742]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator.HARAKAI>nslookup
Default Server: dc.test.net
Address: 192.168.1.2

>
```


Task 1: Enviroment Set Up the Domain Controller (DC)

- Set the static ip for domain controller and point to DNS server to 192.168.1.2



Task 1: Enviroment Set Up the Domain Controller (DC)

1.3 Create Organizational Units (OUs)

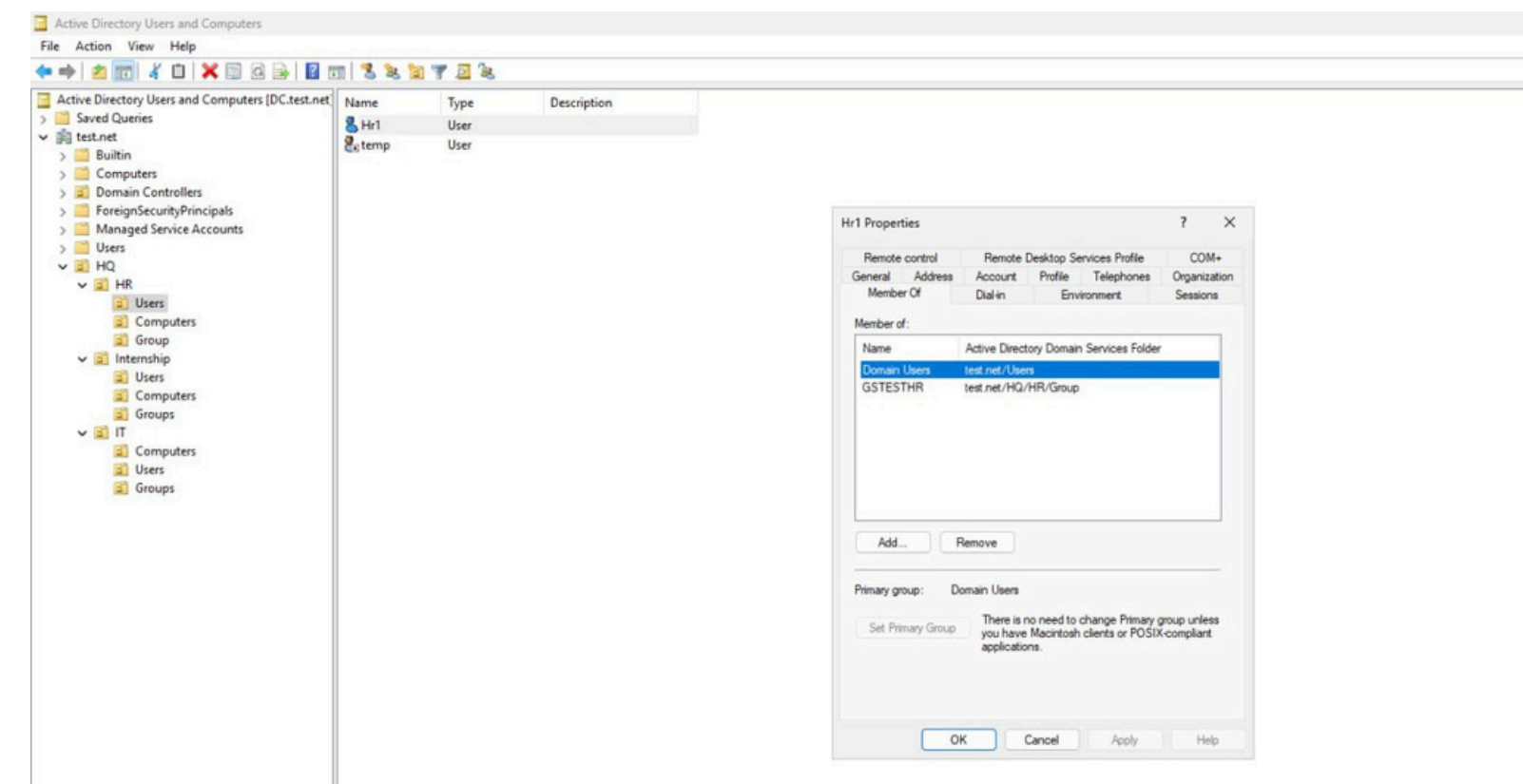
- Open Active Directory Users and Computers (ADUC).
- Right-click domain test.net → New → Organizational Unit:
- Create these OUs: HQ,HR,IT,Internship

1.4 Create User and Computer Containers (inside OUs)

- Under each OU (HQ, HR, IT, Internship), you can create sub-OUs if needed:
- Right-click an OU → New → Organizational Unit → name it Users.
- Right-click an OU → New → Organizational Unit → name it Computers.

1.5 Create Global Security Groups

- In ADUC, right-click HR OU → New → Group:
- Group name: Staff
- Group scope: Global
- Group type: Security
- In Internship OU → New → Group:
- Group name: Intern
- Group scope: Global
- Group type: Security
- Add respective users to these groups.



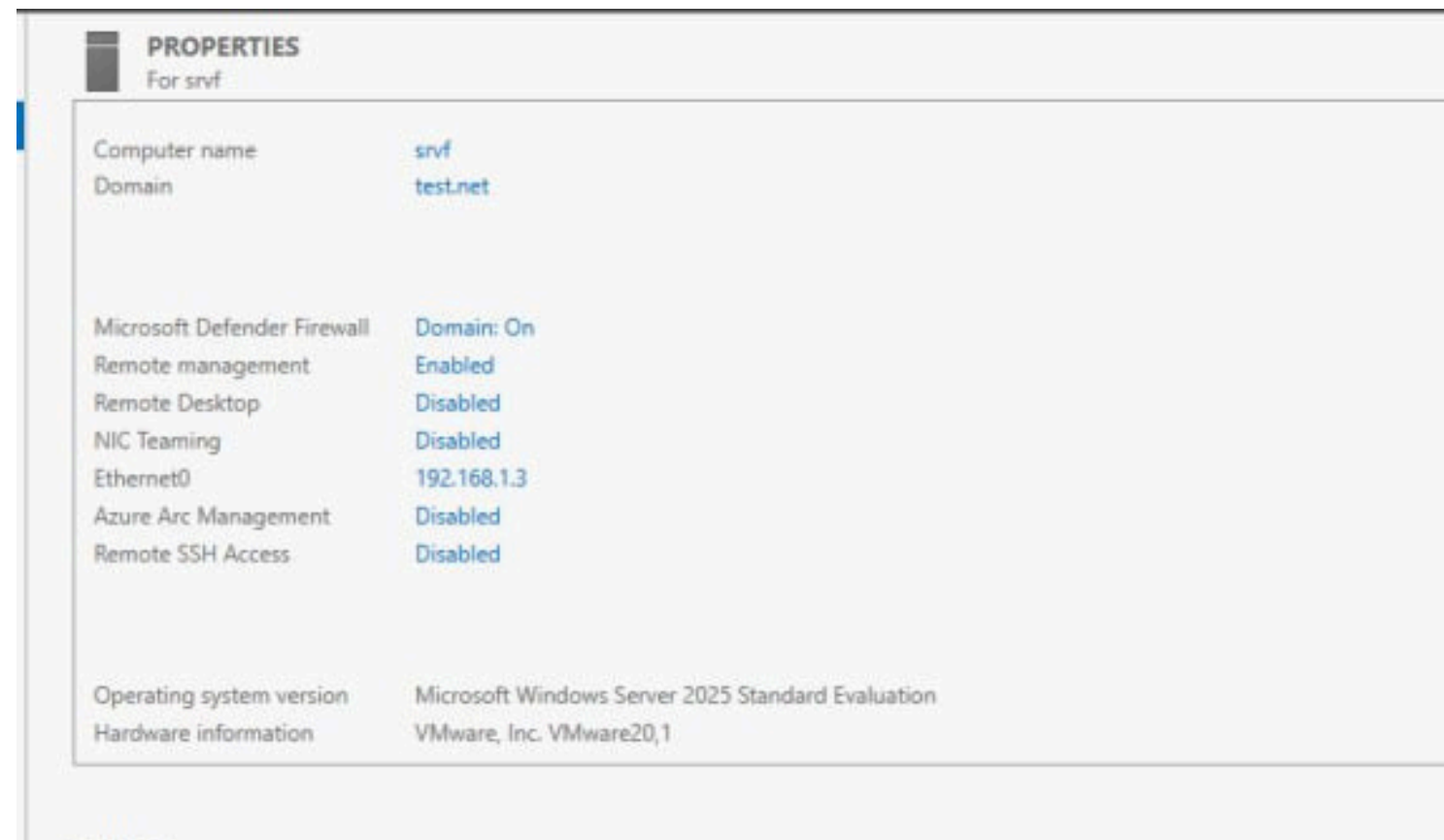
Task 2: Enviroment Set Up FileServer

2.1 Setup File Server (srvf.test.net)

- Install Windows Server.
- Open Server Manager → Add roles and features.
- Install File and Storage Services → File Server role.
- Rename the server to SRVF.
- Set static IP: 192.168.1.3.

2.2 Join File Server to Domain

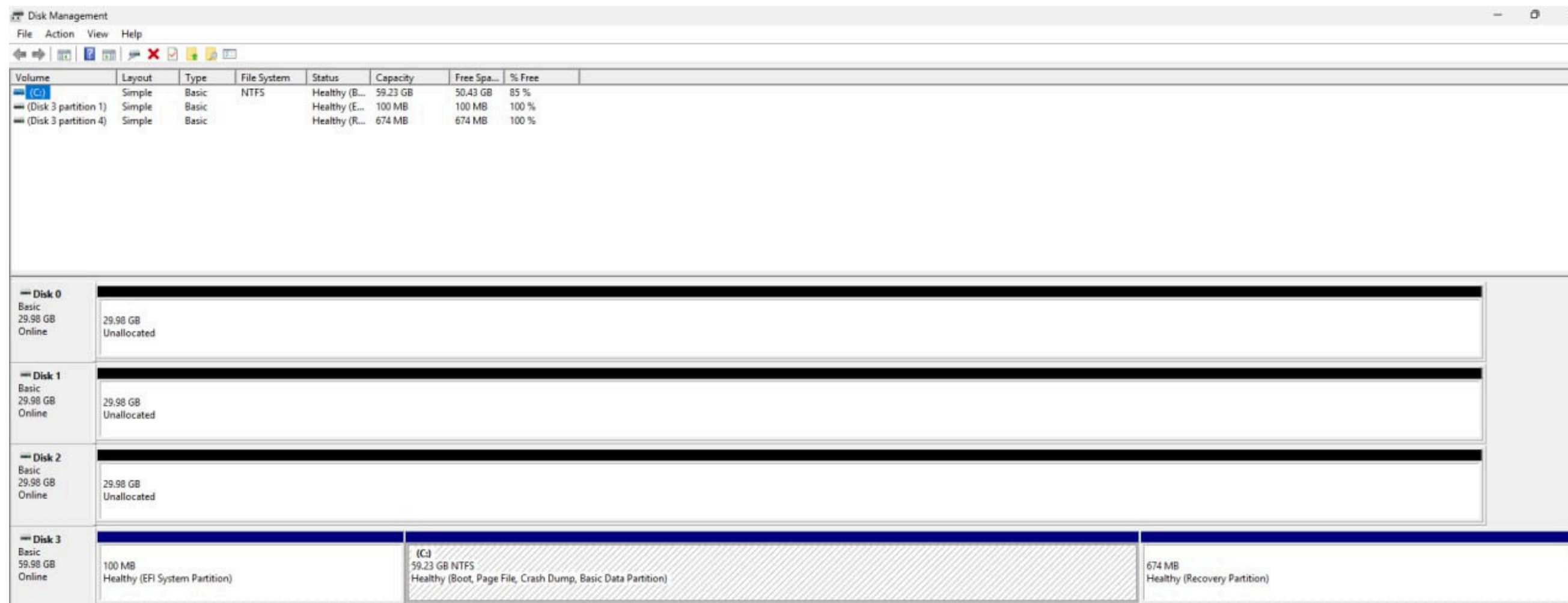
- Go to System Properties → Change settings → Domain: test.net.
- Enter domain admin credentials.
- Restart the server.



Task 2: Enviroment Set Up FileServer

2.3 Configure RAID 5 for HQ Storage

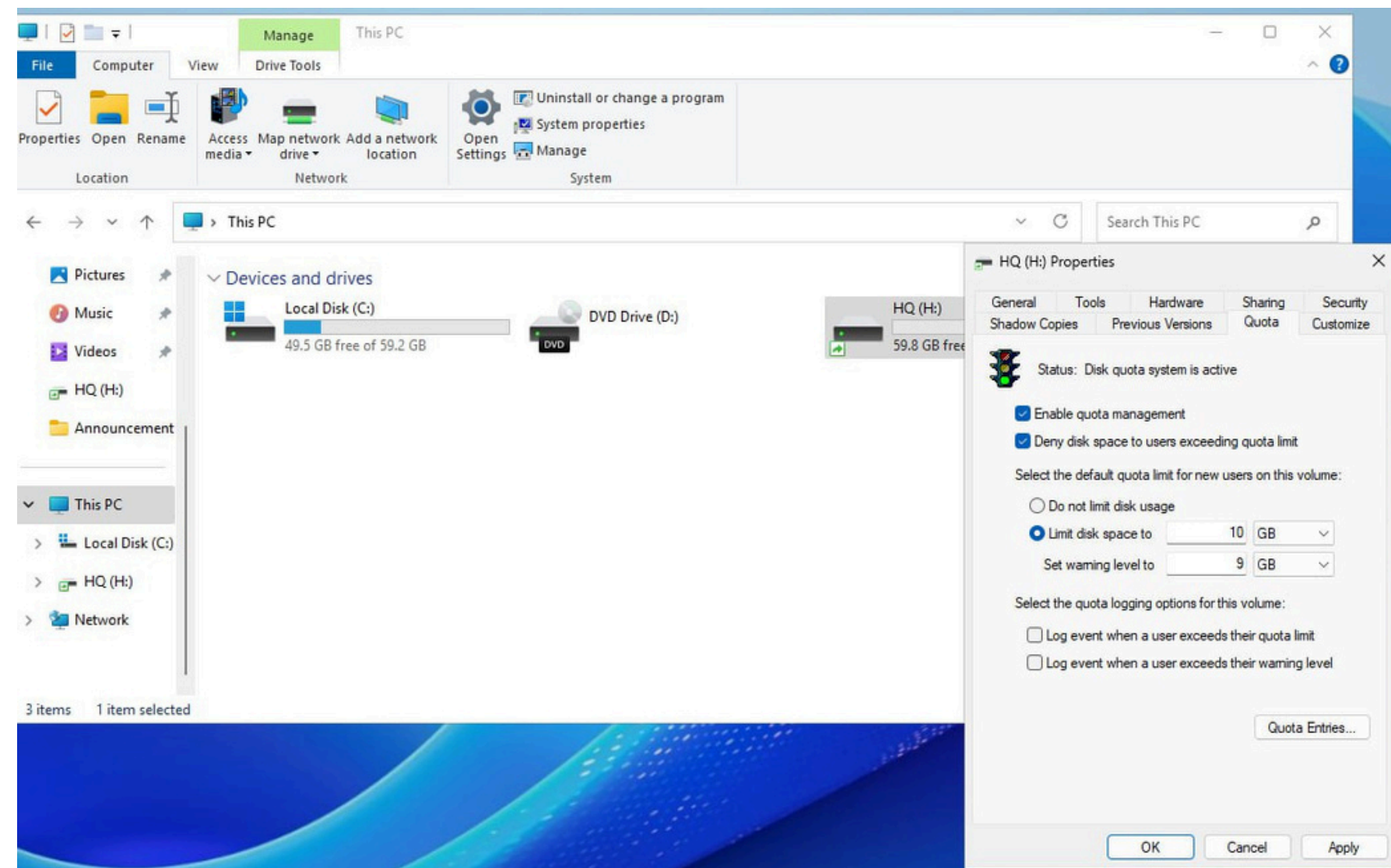
- Disk Management
- Right click on unlocated disk
- Create a Virtual Disk using RAID 5 (3+ disks required).
- Format the virtual disk with NTFS → assign drive letter (e.g., H:) → name it HQ.



Task 2: Enviroment Set Up FileServer

2.4 Set Disk Quota (10 GB per user)

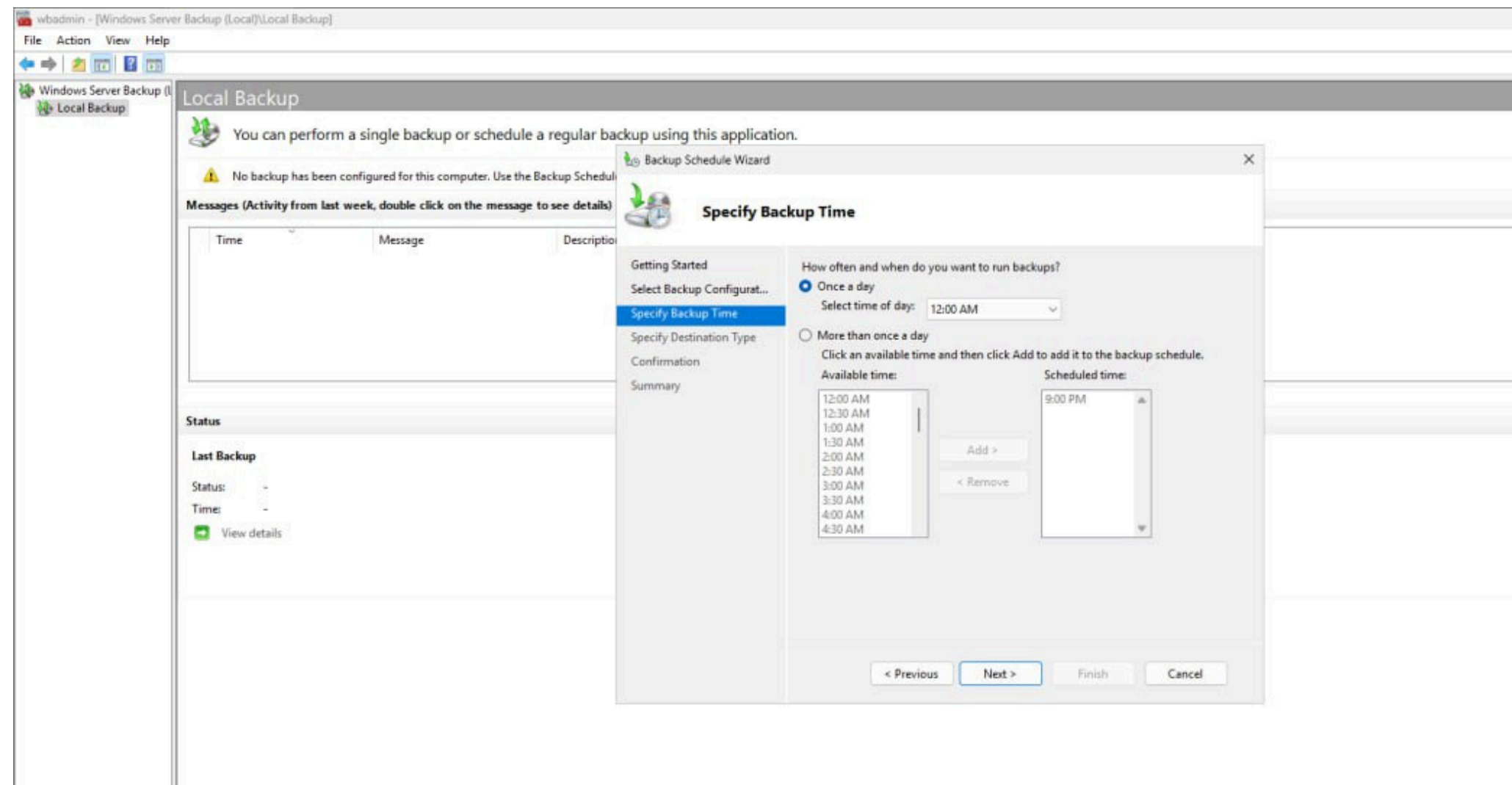
- Right-click the HQ drive (H:) → Properties → Quota tab.
- Enable quota management:
- Check: Enable quota management
- Check: Deny disk space to users exceeding quota
- Set limit: 10 GB
- Set warning level: optional
- Apply settings.



Task 2: Enviroment Set Up FileServer

2.5 Setup Backup Schedule (midnight daily)

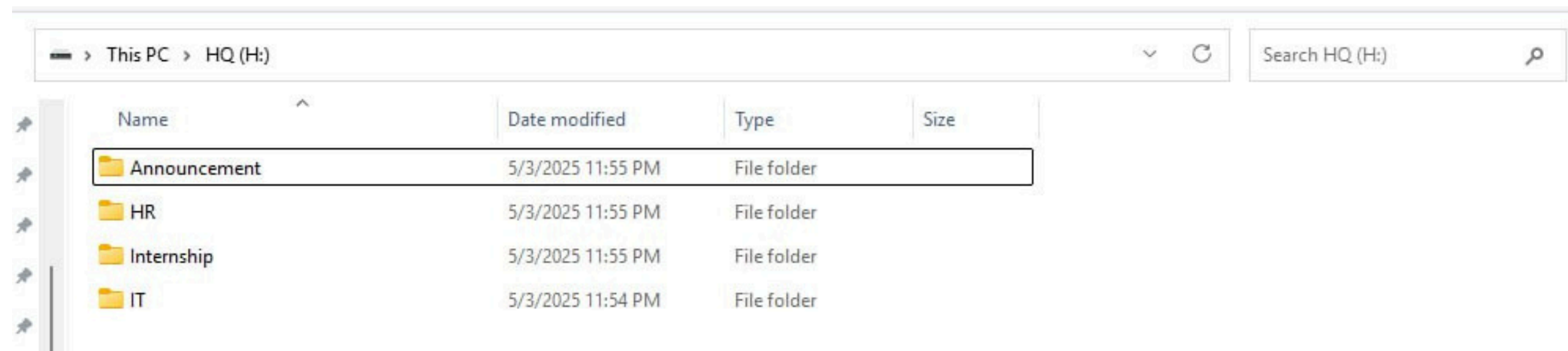
- Open Windows Server Backup → Local Backup.
- Select Backup Schedule → Custom → Add Items → Select folder HQ (H:\HQ).
- Set schedule → Daily at 12:00 AM.
- Choose backup destination (external disk or network location).



Task 2: Enviroment Set Up FileServer

2.6 Create Folders Based on OUs

- In H:\HQ, create folders: HR, IT, Internship, Announcement

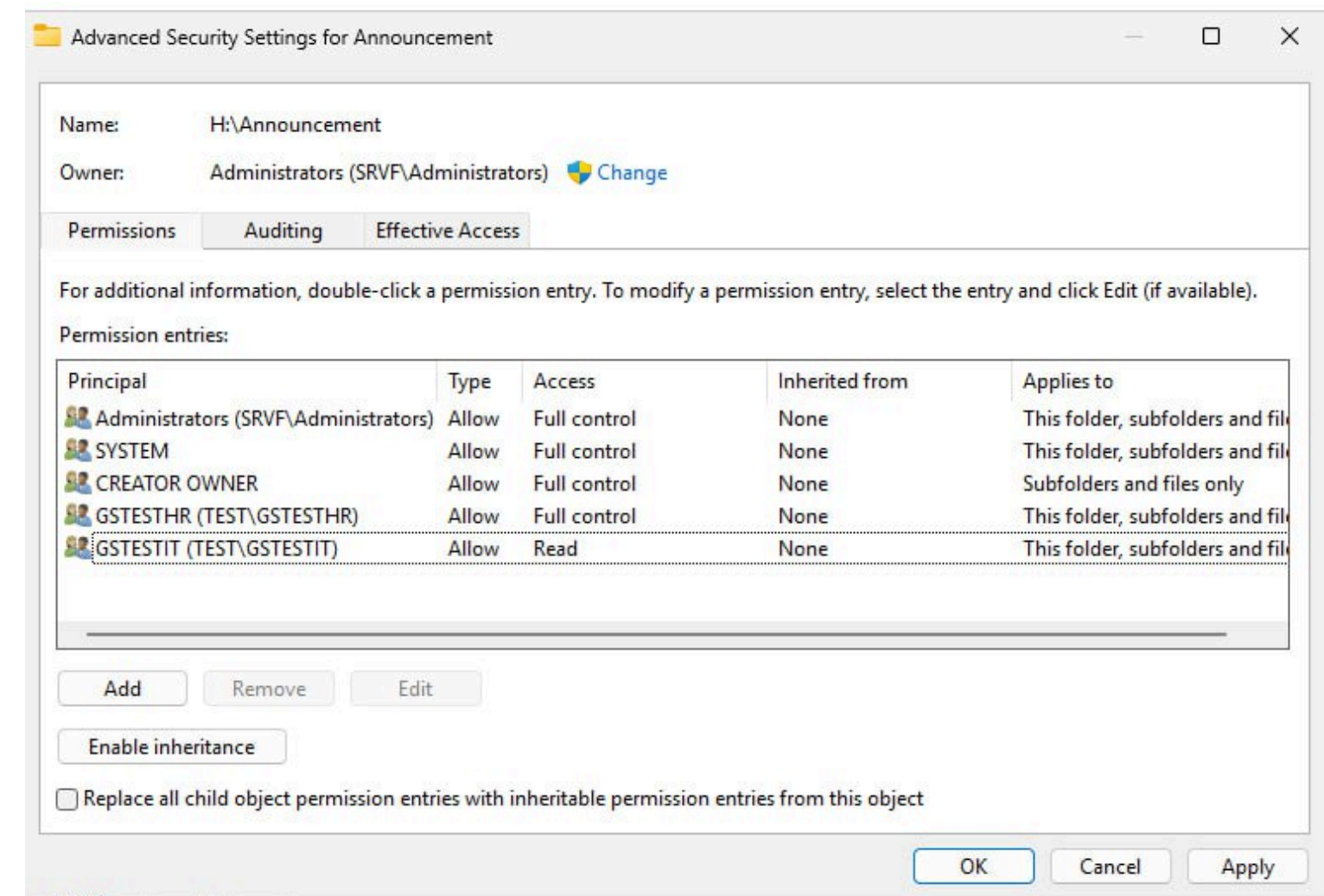
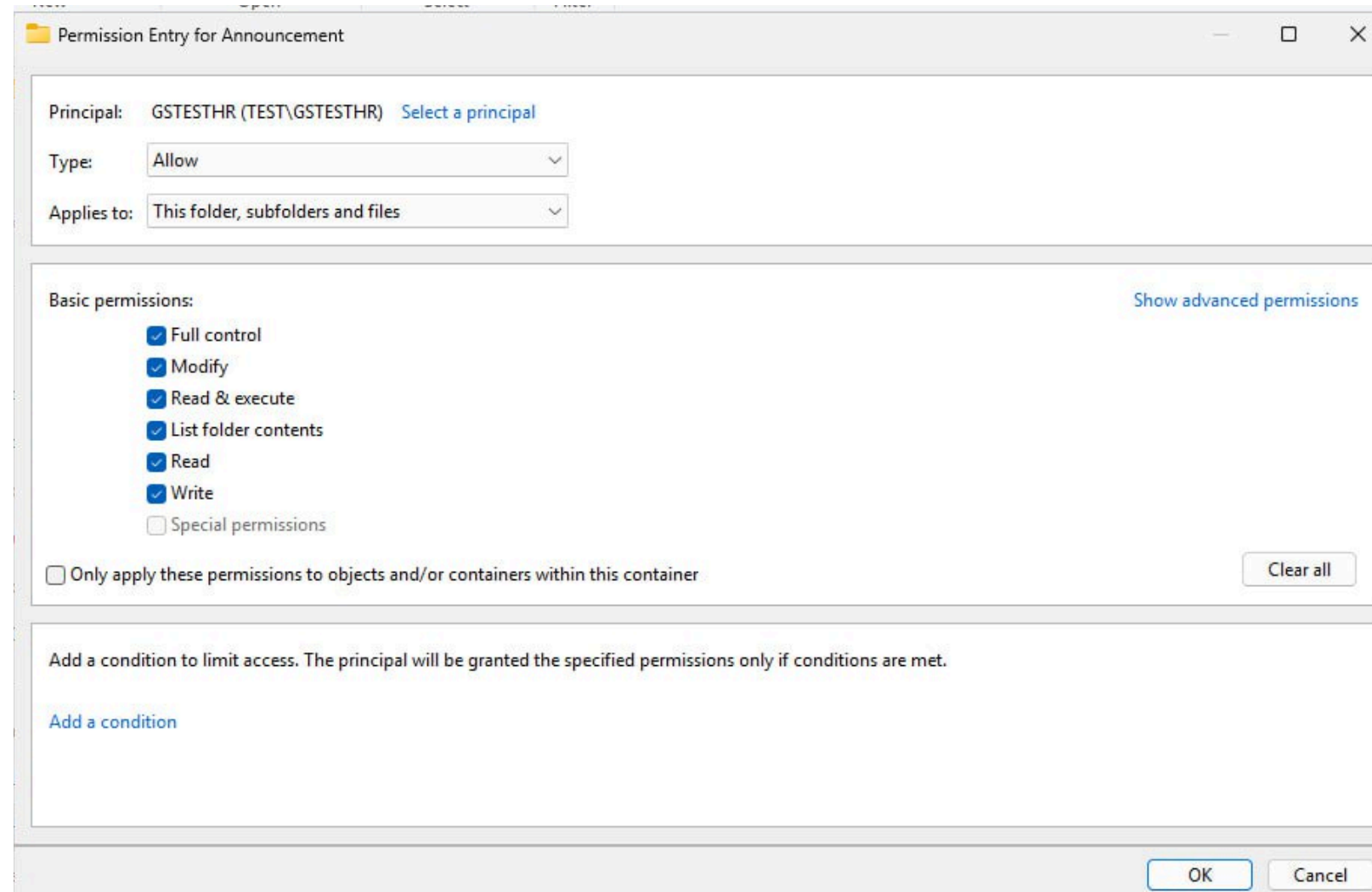


2.7 Configure Permissions for Announcement Folder

- Right-click H:\HQ\Announcement → Properties → Security → Edit.
- Groups/User
 - HR group
 - Staff group
 - Intern group
- Permission
 - Full Control
 - Read
 - Deny Read

Task 2: Enviroment Set Up FileServer

- Setup folder Announcement permission for full control to HR department



Task 2: Enviroment Set Up FileServer

2.8 Set up map drive

2.8.1. Open Group Policy Management on Domain Controller:

- Start → gpmc.msc.
- Right-click domain or OU → Create a GPO → name it: MapDriveH
- Right-click → Edit.

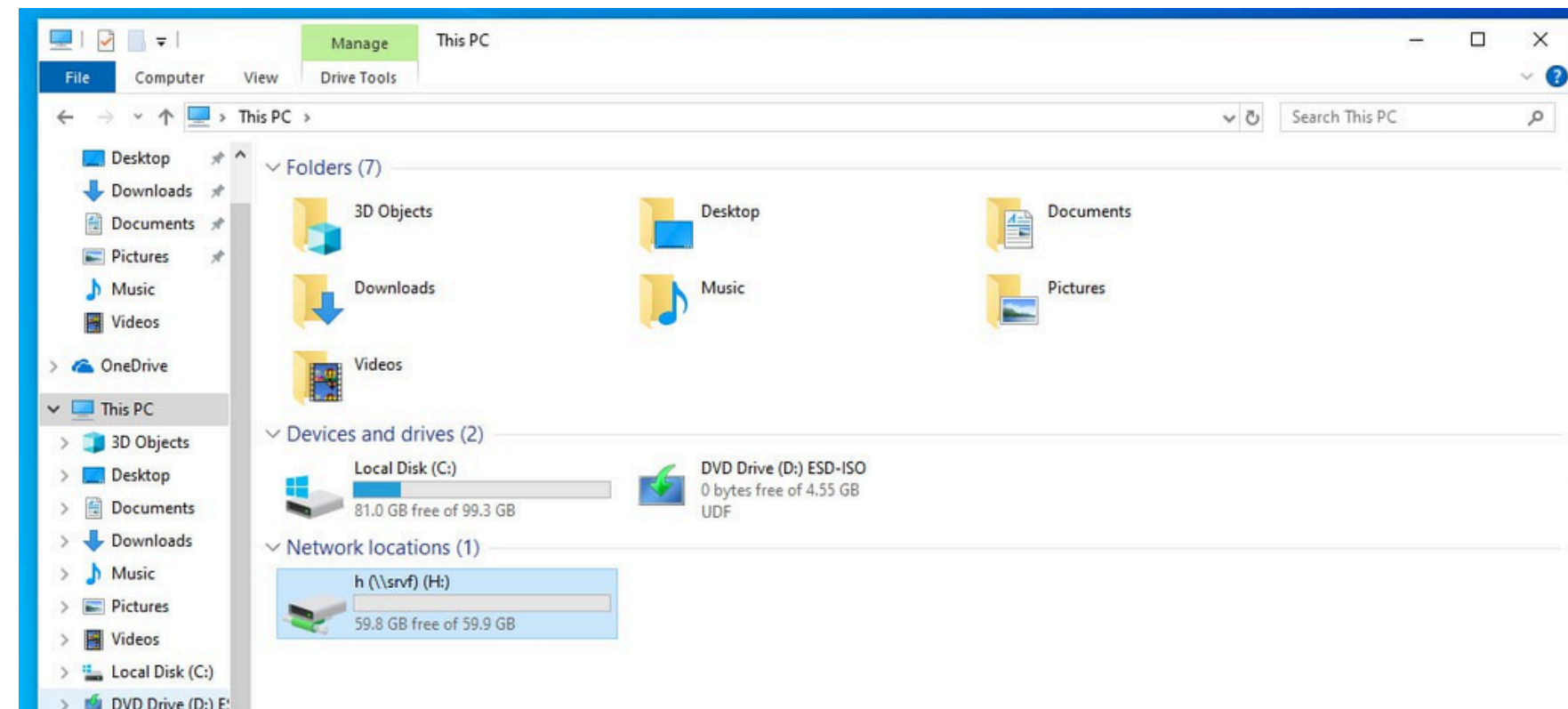
2.8.2. Configure Mapped Drive:

- Inside Group Policy Management Editor:

2.8.2.1 Navigate to:

- User Configuration → Preferences → Windows Settings → Drive Maps
- Right-click → New → Mapped Drive.

Fill the Field:		
Setting		Value:
Action	→	Create
Location	→	\\srvf\h
Label as	→	SRVF
Drive Letter	→	H:



THANK YOU !!!

